Lecture 22 (Zeolites)

Tuesday, 25 April 2023 10:31 AM

K-point density in reciprocal space (higher unit cell requires less k-points)

↳ worse-case (over sampling k-pints for larger unit cell)

↳ Discuss this in the presentation

Solvents in microporous materials

S-L has higher complexity

Modelling accuracy: + Ignoring the solvent
+ Purely thermodynamic (chemical potentials) & phase diagram
+ H-bonding in an average way
+ Static solvent molecules (simulated annealing to find minima energy)
+ AIMD

Water in zeolite beta: how defects affect structure of water in zeolite pore

AIMD simulation
↳ for a few femtoseconds
& then relax w/ DFT

β w/ closed Sn sites → chemi-adsorb water molecules

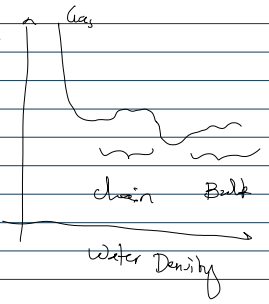
β w/ hydrolized Sn site (OH from water & O from SiO_2) } Si-OH & OH from Sn →

Silanol Nests (OH-β)

AIMD approaches for Entropy: S = Solid-like + Gas-like (Vibrational density of states)

Solid-like Vibrations

Entropy increase upon formation of water chains → Approach to bulk liquid entropy at higher water density



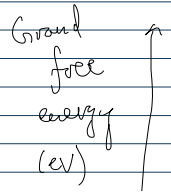
Repeat in calc

Water → gas
→ vibrating solid

Entropy from AIMD
↳ Vib. spectra
↳ we know gas phase (there) & substrate

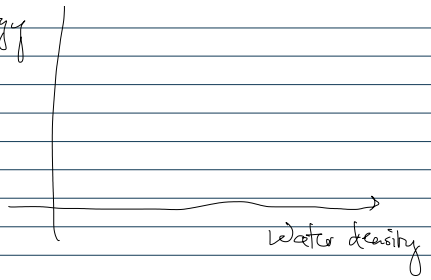
Not long enough to sample all cases (randomly new initialized structures)

↳ Decent values for faster time scales.



↳ Decent values for faster timescales.

↳ Run 10 times as long or average the results over multiple runs.



TST & DFT for Kinetic Modelling

Lower frequency modes contribute the most to entropy ← But also are not accurate

→ Approximate partition function

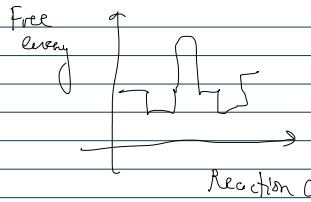
→ Frustrated rotational, translational

→ LA, LB zeolites (Butadiene synthesis

Charlie Campbell

Campbell - Sellers Model for entropy (of weakly bonded adsorbates)

Microkinetic



(Each elementary step barrier is modelled)

Determine RDS

Find rate expression